CONSTRUCTIBILITY IS LOCAL ON A FINITE COVERING BY RETROCOMPACT OPENS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that constructible subsets pull back along an open immersion. Invoked inside the counted proof (source lines 2861-2863). Gives the direct implication.
- The statement that constructible subsets of an open subspace pull back along a quasi-compact open immersion. Invoked inside the counted proof (source lines 2864-2867). Shows each trace is constructible in the ambient space.

Source reference. Original lemma lemma-open-immersion-constructible-inverse-image: sources/stacks-topology/pinned-original-source.tex, source lines 2821–2821. Navigation pointer only: the counted text refers to the Stacks item lemma-open-immersion-constructible-inverse-image, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-open-immersion-constructible-image: sources/stacks-topology/pinned-original-source.tex, source lines 2837–2837. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-open-immersion-constructible-image, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a topological space. Let $E \subset X$ be a subset. Let $X = V_1 \cup \ldots \cup V_m$ be a finite covering by retrocompact opens. Then E is constructible in X if and only if $E \cap V_j$ is constructible in V_j for each $j = 1, \ldots, m$.*

Proof. If E is constructible in X, then by Lemma lemma-open-immersion-constructible-inverse-image we see that $E \cap V_j$ is constructible in V_j for all j. Conversely, suppose that $E \cap V_j$ is constructible in V_j for each $j = 1, \ldots, m$. Then $E = \bigcup E \cap V_j$ is a finite union of constructible sets by Lemma lemma-quasi-compact-open-immersion-constructible-image and hence constructible. □